

MATH 201: Linear Algebra

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Lecture 3 of 12. Eigenvalues and eigenvectors.

Prerequisites: determinants, systems of linear equations, vector spaces.

Definition of an eigenvalue

Let A be a square n by n matrix over the reals. A scalar λ is an eigenvalue of A if there exists a non-zero vector v such that $Av = \lambda v$.

The vector v is called an eigenvector of A corresponding to λ .

The zero vector is never an eigenvector, so an eigenvalue of 0 is still meaningful.

Worked example: for $A = \begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix}$ the characteristic polynomial is $(2 - \lambda)^2$, so $\lambda = 2$ is the only eigenvalue.

Computing the characteristic polynomial

The characteristic polynomial of A is $p(\lambda) = \det(A - \lambda I)$.

Set $p(\lambda) = 0$ and solve for λ over the reals.

The spectrum of A is the set of its eigenvalues, with multiplicity.



Eigenvectors and eigenspaces

For each eigenvalue λ the eigenspace is the null space of $A - \lambda I$.

The geometric multiplicity is $\dim(\ker(A - \lambda I))$ and the algebraic multiplicity is the multiplicity of λ as a root of p .

Diagonalisation of A is possible exactly when every eigenvalue has equal geometric and algebraic multiplicity.

Exam note: questions 3 and 7 of the 2023 paper ask for an eigenspace basis.

Diagonalisation worked example

Take $A = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}$. $\det(A - \lambda I) = (4 - \lambda)(3 - \lambda) - 2 = \lambda^2 - 7\lambda + 10$.

The roots are $\lambda = 5$ and $\lambda = 2$, so A has two distinct eigenvalues.

For $\lambda = 5$ the null space of $A - 5I$ is spanned by $(1, 1)$.

For $\lambda = 2$ the null space of $A - 2I$ is spanned by $(1, -2)$.

Hence $P = \begin{bmatrix} 1 & 1 \\ 1 & -2 \end{bmatrix}$ and $P^{-1}AP = \text{diag}(5, 2)$.